

Here are your notes for the two videos on  $R^2$  and Adjusted  $R^2$ .

# Lecture Notes: $R^2$ and Adjusted $R^2$

## Video 1: R-squared ( $R^2$ ), Clearly Explained!!!

### 1. What is R-squared ( $R^2$ )?

- **R-squared** (or the "coefficient of determination") is a metric that tells you **what proportion of the total variation in your data (Y-axis) is explained by your model** (e.g., a linear regression line).
- It's a value between 0 and 1 (or 0% and 100%).
- $R^2 = 1$  (**100%**): The model is a *perfect fit*. All data points fall exactly on the regression line. The model explains all of the variation.
- $R^2 = 0$  (**0%**): The model is a *useless fit*. The regression line is no better at explaining the data than simply using the average (mean) of all the data points.
- $R^2 = 0.81$  (**81%**): This means that 81% of the total variation in the data is explained by the model, while the remaining 19% is unexplained (residual error).

### 2. How $R^2$ is Calculated

The calculation compares the "error" of your model to the "error" of a baseline, "worst" model.

- **Step 1: Calculate Total Variation (The Baseline)**
  - The "worst model" is just a **horizontal line at the mean (average)** of all your Y-values. This assumes the X-variable has no effect.
  - You calculate the **Sum of Squares around the Mean ( $SS_{mean}$ )**. This is the sum of all the squared vertical distances from each data point to this mean line.
  - This  $SS_{mean}$  represents the **total variation** in the data that you are trying to explain.
- **Step 2: Calculate Unexplained Variation (Your Model)**
  - This is the "error" of your *actual* regression line (the "best fit line").
  - You calculate the **Sum of Squares around the Fit ( $SS_{fit}$ )**, also known as the **Residual Sum of Squares ( $SS_{res}$ )**. This is the sum of all the squared vertical distances from each data point to the regression line.
  - This  $SS_{fit}$  represents the **unexplained variation**—the error your model *doesn't* account for.
- **Step 3: The Formula**
  - The formula for  $R^2$  is:

$$R^2 = 1 - \frac{SS_{fit}}{SS_{mean}}$$

- **Interpretation:**  $R^2 = 1 - \frac{\text{Unexplained Variation}}{\text{Total Variation}}$
- This is the same as:  $R^2 = \frac{\text{Total Variation} - \text{Unexplained Variation}}{\text{Total Variation}} = \frac{\text{Explained Variation}}{\text{Total Variation}}$

### 3. $R^2$ vs. Correlation ( $r$ )

- $R^2$  is literally the **square of Pearson's correlation coefficient ( $r$ )**.
- $R^2 = r^2$
- If your correlation  $r = 0.7$ , your  $R^2 = (0.7)^2 = 0.49$ . This means that a correlation of 0.7 *only explains 49%* of the variation in the data.
- This makes  $R^2$  a more intuitive measure of a model's explanatory power.

# Video 2: Adjusted R-Squared, Clearly Explained!!!

## 1. The Problem with $R^2$

- $R^2$  has a major flaw: **it will *always* increase (or stay the same) every time you add a new variable** (predictor) to your model.
- It *never* goes down.
- This is true even if the new variable you add is **complete junk** (e.g., random numbers). The model will "overfit" and use that junk variable to explain a tiny, random portion of the noise, which artificially inflates the  $R^2$ .
- This makes  $R^2$  useless for comparing models with *different numbers of variables*.

## 2. The Solution: Adjusted $R^2$

- **Adjusted  $R^2$**  is a modified version that **penalizes the model for adding variables that don't add real predictive power**.
- It only increases if the new variable improves the model *more than would be expected by chance*.
- If you add a junk variable, your Adjusted  $R^2$  will **go down**.
- This makes it the correct tool for comparing models with different numbers of predictors (e.g., deciding if you should add a variable to your model).

## 3. The Formula (and its penalty)

The formula looks more complex, but the new parts are just the "penalty":

$$R_{adj}^2 = 1 - \frac{SS_{fit}/(n - p - 1)}{SS_{mean}/(n - 1)}$$

- $SS_{fit}$ : Unexplained variation (same as before).
- $SS_{mean}$ : Total variation (same as before).
- $n$ : The sample size (number of data points).
- $p$ : The **number of predictors** (variables) in your model.

## 4. How the Penalty Works

- The denominator  $SS_{mean}/(n - 1)$  is just the **variance of the Y-values**. For a given dataset, this value is **constant**.
- The numerator  $SS_{fit}/(n - p - 1)$  is the **Residual Mean Square**. The key is its denominator:  $(n - p - 1)$ .
- When you **add a new variable**,  $p$  increases by 1.
- This makes the term  $(n - p - 1)$  **get smaller**.
- Dividing  $SS_{fit}$  by a *smaller* number makes the entire numerator fraction **get larger**.
- Subtracting a *larger* fraction from 1 makes the final  $R_{adj}^2$  **get smaller**.

**In short:** Adding a variable (increasing  $p$ ) automatically applies a penalty that *lowers* the Adjusted  $R^2$ . For the Adjusted  $R^2$  to *increase*, the new variable must be so good that it *slashes* the  $SS_{fit}$  (unexplained error) by enough to overcome that penalty. A junk variable can't do this.

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